

PART 1: Conditional probability graphs

We can use the full joint probability tables to answer any question we may have about a set of random variables; however, as the number of variables increases, the size of the joint probability table increases exponentially. Thankfully, we can use the independence and conditional independence relationships between variables to design a compact representation that greatly reduces the number of probabilities needed to define the full joint probability distribution. These structures are called *Bayesian networks* and are typically represented as directed, acyclic graphs $G = (V, E)$ where V is the set of random variables and $E \subseteq V \times V$ is the set of directed edges. Random variables are denoted by capital letters (e.g., X) and lower-case denotes the truth assignment of a variable (e.g., x or $\neg x$). Edge $(Y, X) \in E$ indicates that Y is the *parent* of X and we can denote the set of parents of X as $parents(X)$. The parents of X are the only variables that *directly* affect the probability of X . This means that given $parents(X)$, there is no other information that is useful to estimate $P(X)$, formally: $P(X | parents(X)) = P(X | V_1, V_2, \dots, V_n)$ for all random variables $V_i \in V$.

In order to compute the full joint distribution using a Bayesian network, we need to find the product of all the conditional probabilities. We can do this using the following equation:

$$P(x_1, x_2, \dots, x_n) = \prod_{i=1}^n P(x_i | parents(X_i)).$$

It is clear that if X_i has no parents, its “conditional” probability is simply $P(X_i)$.

Where we are making a query about a variable but we don’t have full knowledge about its parents, we must sum over the hidden variables. This can be done in the following manner:

$$P(X) = \sum_{\mathbf{y}} P(X, \mathbf{y}),$$

This technique is known as *marginalisation* and here, $\mathbf{Y}$ is the vector of all hidden variables we wish to sum over. For example, in the case where there is one hidden variable Y , we have,

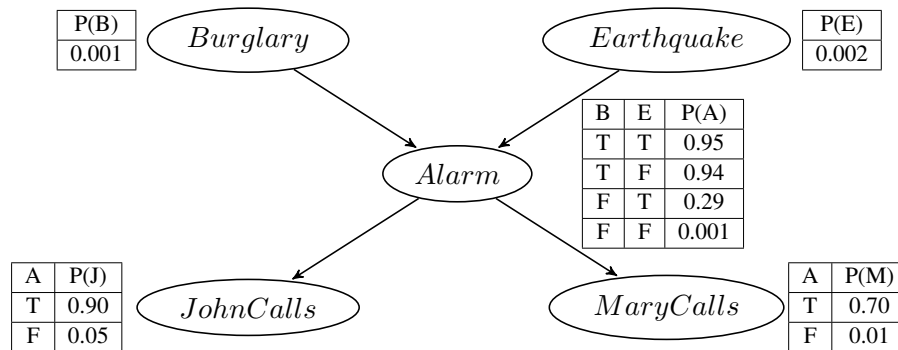
$$P(X) = P(X, y) + P(X, \neg y);$$

in the case of two hidden variables Y_1 and Y_2 we have,

$$P(X) = P(X, y_1, y_2) + P(X, y_1, \neg y_2) + P(X, \neg y_1, y_2) + P(X, \neg y_1, \neg y_2);$$

and so on. In general, we will have the sum of 2^n terms, where n is the number of hidden variables being summed over. Each of these terms will represent one of the possible truth assignments for all of the hidden variables.

Example: Consider the following Bayesian network:



- (a) Give an expression for the probability $P(j, m, a, \neg e, \neg b)$.

Solution: Here, we need to find conditional probabilities for each truth assignment of the variables, based on the structure of the graphical model given. So, j depends on a ; m depends on a ; a depends on both e and b ; and, b and e don't depend on any other variables. The total probability for the entry in the full joint table represented by $P(j, m, a, \neg e, \neg b)$ is the product of all of these conditional probabilities.

Therefore an expression for the above probability would be:

$$P(j | a) \times P(m | a) \times P(a | \neg e, \neg b) \times P(\neg e) \times P(\neg b)$$

- (b) Compute the probability $P(j, m, a, \neg e, \neg b)$.

Solution: Having worked out an expression for the probability in the previous question, computing the value is a simple matter of plugging in the correct values - remembering that $P(\neg a) = 1 - P(a)$.

Therefore, the value for the above probability is:

$$\begin{aligned}
 P(j, m, a, \neg e, \neg b) &= P(j | a) \times P(m | a) \times P(a | \neg e, \neg b) \times P(\neg e) \times P(\neg b) \\
 &= P(j | a) \times P(m | a) \times P(a | \neg e, \neg b) \times (1 - P(e)) \times (1 - P(b)) \\
 &= 0.90 \times 0.70 \times 0.001 \times (1 - 0.002) \times (1 - 0.001) \\
 &= 0.90 \times 0.70 \times 0.001 \times 0.998 \times 0.999 \\
 &= 0.0006281113
 \end{aligned}$$

- (c) Give an expression for the probability $P(j, \neg m, \neg a, b)$.

Solution: Here, we are missing one of the truth assignment to the variable e , so we must marginalise over that variable by summing up all entries in the full joint table corresponding to all of the other truth assignments and when e is both true *and* when it is false.

Therefore, an expression for the above probability is:

$$\begin{aligned}
 P(j, \neg m, \neg a, b) &= \sum_e P(j, \neg m, \neg a, b, e) \\
 &= \sum_e P(j | \neg a) \times P(\neg m | \neg a) \times P(\neg a | b, e) \times P(b) \times P(e) \\
 &= P(j | \neg a) \times P(\neg m | \neg a) \times P(\neg a | b, e) \times P(b) \times P(e) \\
 &\quad + P(j | \neg a) \times P(\neg m | \neg a) \times P(\neg a | b, \neg e) \times P(b) \times P(\neg e)
 \end{aligned}$$

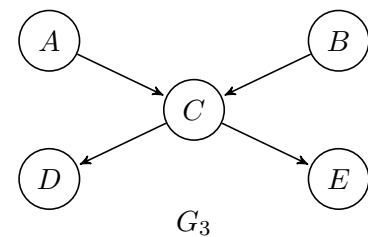
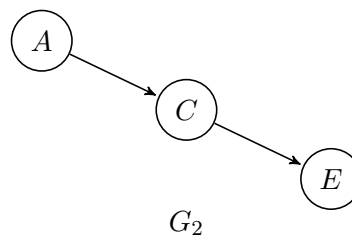
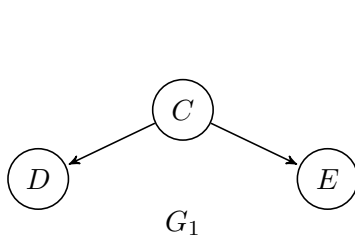
- (d) Compute the probability $P(j, \neg m, \neg a, b)$.

Solution: Just plug the values in as you did before:

$$\begin{aligned}
 P(j, \neg m, \neg a, b) &= P(j \mid \neg a) \times (1 - P(m \mid \neg a)) \times (1 - P(a \mid b, e)) \times P(b) \times P(e) \\
 &\quad + P(j \mid \neg a) \times (1 - P(m \mid \neg a)) \times (1 - P(a \mid b, \neg e)) \times P(b) \times (1 - P(e)) \\
 &= 0.05 \times 0.99 \times 0.05 \times 0.001 \times 0.002 \\
 &\quad + 0.05 \times 0.99 \times 0.06 \times 0.001 \times 0.998 \\
 &= 0.00000296901
 \end{aligned}$$

Questions:

Consider the following directed graphical models:



Give expressions for the following probabilities using only terms of the form $P(X \mid \text{parents}(X))$:

(a) with respect to graph G_1 ;

- i. $P(c, d, e)$
- ii. $P(d, e \mid c)$

(b) with respect to graph G_2 ;

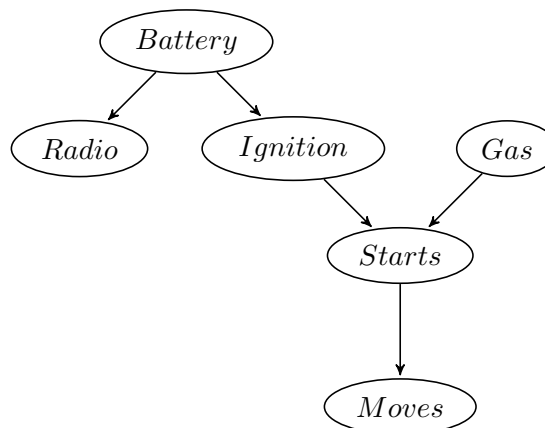
- i. $P(a, c, e)$
- ii. $P(a, e \mid c)$

(c) with respect to graph G_3 .

- i. $P(a, \neg b, c, d, \neg e)$
- ii. $P(a \mid \neg d, e)$

PART 2: Car diagnosis

Consider the network for car diagnosis shown in the figure below:



- (a) Extend the network with the Boolean variables *IcyWeather* and *StarterMotor*, by reasoning how components might affect each other. Being a design/modelling process, there may be many correct answers, just state your assumptions.

- (b) Give reasonable conditional probability tables for all the nodes.
- (c) How many independent values are contained in the joint probability distribution for eight Boolean nodes, assuming that no conditional independence relations are known to hold among them?
- (d) How many independent probability values do your network tables contain?
- (e) **(optional)** The conditional distribution for *Starts* could be described as a **noisy-AND** distribution. Define this family in general and relate it to the **noisy-OR** distribution.
- (f) Reconstruct, showing your workings, five entries of the full joint probability distribution (e.g., reconstruct $P(\neg iW, b, r, i, g, s, sM, \neg m)$ or $P(iW, \neg b, r, i, \neg g, \neg s, sM, m)$, or any other variable assignment).

PART 3: Power station diagnosis

In your local nuclear power station, there is an alarm that senses when a temperature gauge exceeds a given threshold. The gauge measures the temperature of the core. Consider the Boolean variable A (alarm sounds), F_A (alarm is faulty), and F_G (gauge is faulty) and the multivalued nodes G (gauge reading) and T (actual core temperature).

- (a) Draw a Bayesian network for this domain, given that the gauge is more likely to fail when the core temperature gets too high.
- (b) Is your network a polytree?
- (c) Suppose there are just two possible actual and measured temperatures, normal and high; the probability that the gauge gives the correct temperature is x when it is working, but y when it is faulty. Give the conditional probability table associated with G .
- (d) Suppose the alarm works correctly unless it is faulty, in which case it never sounds. Give the conditional probability table associated with A .
- (e) **(optional)** Suppose the alarm and gauge are working and the alarm sounds. Using the information from the previous questions, calculate an expression for the probability that the temperature of the core is too high, in terms of the various conditional probabilities in the network.

PART 4: Bayesian Networks via pgmpy

Finally, let's get our hands dirty with Bayesian Networks in Python!

One can build and do inference with Bayesian Networks in Python using the **pgmpy** package!

So, head to **pgmpy** home page and:

- (a) Install and set-up the package in your machine.
- (b) Understand and run the **Earthquake example** in the documentation.
- (c) Use the system to perform and validate some of the inferences you did in Part 1 for this domain.
- (d) **Enjoy & learn Bayesian Networks with Python!** Make sure you understand what the code is doing and that it does not appear as “magic” to you!

End of tutorial worksheet